

An Overview of Smart Irrigation Systems Using IoT

This paper is a review paper on smart irrigation systems using IoT. The main purpose of the paper is to explain how IoT, sensors, wireless communication, cloud systems, and automation can improve irrigation in agriculture. The paper focuses on saving water, improving soil and weather monitoring, and supporting sustainable farming. It explains that traditional irrigation often wastes water, but smart irrigation can provide water only when the crop actually needs it.

Since this is a review paper, it does not use one fixed hardware prototype. However, it discusses the important components and technologies used in smart irrigation systems. These include soil moisture sensors, temperature sensors, humidity sensors, weather sensors, wireless sensor networks, microcontrollers, Wi-Fi, GSM, LoRa, MQTT, cloud storage, mobile applications, pumps, valves, artificial intelligence, machine learning, and fuzzy logic. Soil moisture sensors are used to measure water level in the soil. Weather sensors collect environmental data. Communication modules send the data to the cloud. Pumps and valves control the water supply.

The paper includes a smart irrigation system diagram. The diagram shows the main parts of a smart irrigation system: data acquisition through sensors, irrigation control, wireless communication, data processing, and fault detection. Another diagram explains the structure of smart irrigation, including sensor nodes, cloud server, mobile application, and irrigation control system.

The smart irrigation system works by collecting soil and weather data from the field. First, the soil moisture sensor checks whether the soil is dry or wet. Weather sensors collect data such as temperature, humidity, wind, rainfall, and sunlight. These data are sent to a controller or cloud platform through wireless communication. The system analyzes the data and decides whether irrigation is needed. If the soil moisture is below the required level, the pump or valve is turned on. If the soil already has enough water, irrigation is stopped. This process reduces water waste and improves crop growth.

This paper is directly aligned with our AI-IoT smart farming rover project because irrigation is one of the major parts of smart farming. Our rover can move through different areas of the field and measure soil moisture from multiple points. Instead of using fixed sensors in only one place, the rover can collect data from different locations. Based on this data, the farmer can know which area needs water. In the future, the rover can also control a pump or send irrigation recommendations to a dashboard. Therefore, this paper supports the soil monitoring, water-saving, and smart irrigation decision-making part of our project.

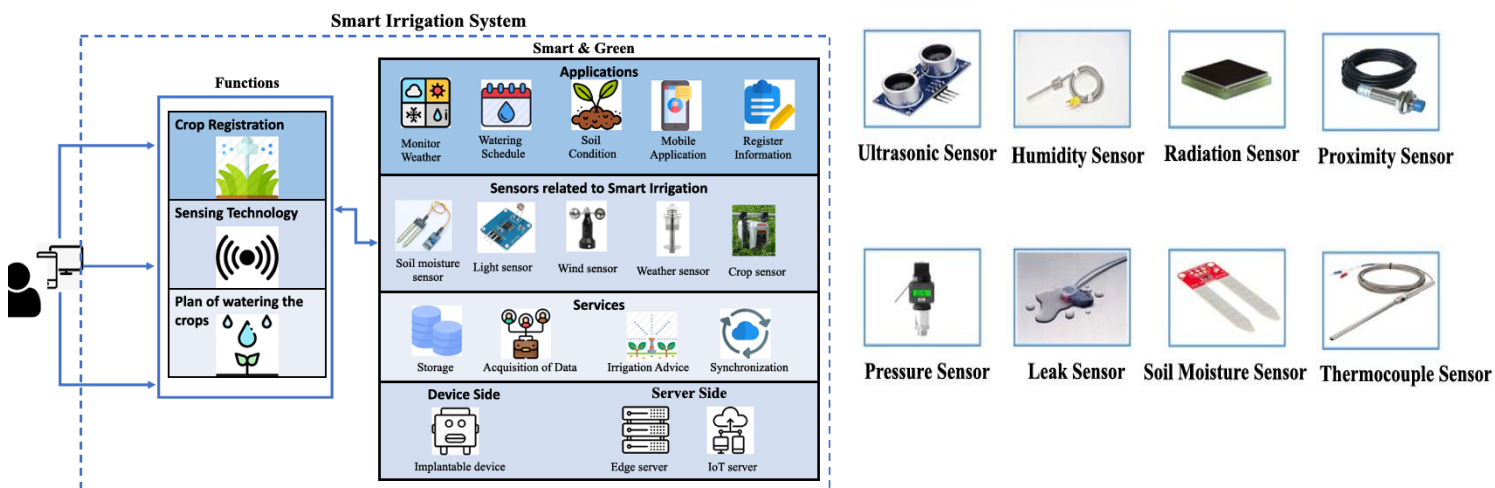


Fig. 6. Smart irrigation system structures, icons taken from flaticon.com.